OPTIMAL WINDOWED TEST FUNCTIONS FOR BOUNDING ANALYTIC RANKS OF ELLIPTIC CURVES

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Abstract. Assuming the Generalized Riemann Hypothesis, the explicit formula for the L-function of an elliptic curve $E/\mathbb{Q}$ bounds the analytic rank of E in terms of the traces a_p for $p < e^{2L}$, for any admissible test function supported on a window $[-L, L]$. In the literature this test function is fixed in advance, typically to a Fejér-type kernel. We instead choose, for each individual curve, the test function that minimises the resulting bound within a finite-dimensional space supported on $[-L, L]$. The minimiser is the solution of a linear system with the Gram matrix of the Weil quadratic form of E, and the optimal bound equals the reciprocal of the corresponding reproducing kernel at the central point, which we call the *radix Riemanniana* of E. Because the Fejér-type kernel lies in this space, the optimised bound is never weaker. We certify every bound with ball arithmetic, check the prime-side evaluation against the zero side computed independently from the zeros of $L(E, s)$, and benchmark against the fixed kernel on 1642 curves from Cremona's tables, 2500 curves from the Stein–Watkins database and a set of curves with conductors up to $3 \cdot 10^{16}$. Depending on the sample and on the stopping criterion, the number of primes required to pin down the analytic rank decreases by 8–23% in aggregate. For individual curves of large conductor the decrease ranges from 0% to 43%. The gain is largest where the method is most expensive: on the curves of rank 28–31 in the ICARM leaderboard the optimised kernel needs 84–85% fewer primes to certify the rank, and for the rank-30 record curve it certifies the bound with $1.35 \cdot 10^7$ primes, while the fixed kernel does not reach it with all $5.08 \cdot 10^7$ primes below 10^9. We validate the fixed-kernel baseline by reproducing every published value of Bober and of Klagsbrun, Sherman and Weigandt with an independent implementation.

1. Introduction

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N and let $r_{\mathrm{an}}(E)$ be the order of vanishing of $L(E, s)$ at the central point. Under the Generalized Riemann Hypothesis (GRH), Mestre's explicit-formula method [10] bounds $r_{\mathrm{an}}(E)$ from above by a quantity computed from the traces of Frobenius a_p for p below a cutoff. Bober [2] made this practical with a Fejér-type test function, and Klagsbrun, Sherman and Weigandt [8] used the same function to show that the Elkies curve has rank 28 subject only to GRH. In these works the shape of the test function is fixed, and only its support (the cutoff) is varied.

This note asks a simple question: *for a given curve and a given cutoff, which test function gives the best bound?* Within any finite-dimensional space of admissible functions supported on $[-L, L]$ the answer is explicit. The explicit formula expresses the bound as a ratio of a quadratic form (the Weil form of E, evaluated through the a_p) to the square of a linear functional. The minimiser is therefore obtained from one linear solve with a Gram matrix. The optimal value is the reciprocal of the reproducing kernel at 0 of the space equipped with the Weil inner product.

What is known, and what is new. The Weil quadratic form restricted to functions supported on a finite interval, and its Gram matrices with respect to a basis, have been studied for the

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Riemann zeta function by Yoshida [15], by Connes, Consani and Moscovici [4], by Suzuki [14] via screw functions and de Branges spaces, and recently by Shi [12] as finite "Prime–Weil matrices". Reproducing-kernel methods for Fourier optimisation problems attached to *families* of L-functions, where the input is the limiting one-level density rather than data of an individual L-function, are due to Carneiro, Chirre and Milinovich [3]. To the best of our knowledge, which rests on a literature search that is not exhaustive, the optimisation of the test function *for an individual elliptic curve, using its actual* a_p, with certified output and a systematic comparison against the fixed kernel, has not been carried out before. That optimisation is the contribution of this note. The underlying theory is classical. The contribution is the per-curve optimisation, its certification, and the measurements.

Results in brief.

- The optimised bound is never weaker than the fixed-kernel bound at the same cutoff (Proposition 3.4). This is a consequence of the construction, not an empirical finding. The empirical question is *how much* it gains.
- Every reported bound is certified with ball arithmetic. No certified bound in our tests falls below the known rank.
- The prime-side value of the Weil form agrees with the zero-side value computed from the zeros of $L(E,s)$. The discrepancy halves with each doubling of the height up to which zeros are included (Section 5).
- The number of primes needed to determine the analytic rank (Section 6) decreases in aggregate by 8.3–15.2% (Cremona, $N < 4.1 \cdot 10^5$) and by 17.0–22.7% (Stein–Watkins, $4.4 \cdot 10^6 \le N \le 7.5 \cdot 10^9$), depending on the stopping criterion. For individual curves of conductor up to $5 \cdot 10^{15}$ the decrease ranges from 0% to 43%.
- On curves of record rank (Section 8) the gain is much larger. For the smallest-conductor curves of ranks 28–31 in the ICARM leaderboard, the number of primes needed for a bound below $r+2$ drops by a factor of about 6.5. For the bound below $r+1$ the fixed kernel fails for three of the four curves with all primes below 10^9, while the optimised kernel succeeds with between $2.3 \cdot 10^5$ and $1.3 \cdot 10^7$ primes. On the historic record curves E_{20}–E_{27} the factor is 2–3.5.
- The fixed-kernel baseline is exactly Bober's method: an independent implementation of Bober's formula reproduces all seven published values in [2, 8] and agrees with our certified evaluation to 10^{-12} (Section 7).

2. The explicit formula and rank bounds

Write $L(E,s) = \sum_{n\ge1} a_n n^{-s-1/2}$ so that the central point is $s = 1/2$, and put

$$\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\,\Gamma(s+\tfrac12)\,L(E,s), \qquad \Lambda(E,s) = \pm\Lambda(E,1-s).$$

For $p \nmid N$ let α_p, β_p be defined by $\alpha_p + \beta_p = a_p/\sqrt{p}$, $\alpha_p\beta_p = 1$, and set $c_{p^k} = \alpha_p^k + \beta_p^k$. For $p \mid N$ set $c_{p^k} = (a_p/\sqrt{p})^k$ (so $c_{p^k} = 0$ for additive reduction). Then $-\frac{L'}{L}(E,s) = \sum_{n\ge2} \Lambda(n) c_n n^{-s}$ in the normalisation above.

Let $f\colon \mathbb{R} \to \mathbb{R}$ be even, bounded and supported in $[-L, L]$. Put $g = f * f$, so that g is even and supported in $[-2L, 2L]$, and

$$\hat f(r) = \int_{\mathbb{R}} f(x) e^{irx}\,dx, \qquad h(r) = \hat g(r) = \hat f(r)^2.$$

Writing the nontrivial zeros as $\rho = \frac{1}{2} + i\gamma$ (with multiplicity), the explicit formula reads

$$(1)\quad \sum_{\gamma} h(\gamma) \;=\; g(0)\log\frac{N}{4\pi^2} \;+\; 2\Big[-\gamma_{\mathrm{E}}\, g(0) + \int_0^\infty \frac{g(0)-g(u)}{e^u-1}\,du\Big] \;-\; 2\sum_{n\ge 2}\frac{\Lambda(n)\,c_n}{\sqrt{n}}\,g(\log n),$$

where γ_{E} is Euler's constant. Only prime powers $n < e^{2L}$ contribute. We denote the right-hand side by $W_E(f)$.

Lemma 2.1. *Assume GRH for $L(E,s)$. Then for every nonzero f as above,*

$$r_{\mathrm{an}}(E) \;\le\; \frac{W_E(f)}{\hat f(0)^2}.$$

Proof. Under GRH every γ is real. Since f is real and even, $\hat f$ is real, so $h(\gamma) = \hat f(\gamma)^2 \ge 0$. The central zero contributes $r_{\mathrm{an}}(E)\,\hat f(0)^2$ to the left-hand side of (1), and every other term is nonnegative. □

Remark 2.2. For $f = \mathbf{1}_{[-L,L]}$ one has $g(u) = (2L-|u|)_+$ and $h(r) = 4\sin^2(Lr)/r^2$. Up to scaling, this is the Fejér-type test function used in [2, 8]. We call the resulting bound $B_{\mathrm{std}}(L)$. With $L = \pi\Delta$ it coincides with Bober's $\sum_\gamma f(\gamma;\Delta)$: $\hat f(r)^2/\hat f(0)^2 = (\sin(\Delta\pi r)/(\Delta\pi r))^2$, the conductor term becomes $\log(\sqrt{N}/2\pi)/(\pi\Delta)$, and the archimedean term evaluates to Spicer's closed form $-\gamma_{\mathrm{E}}/(\pi\Delta) + \big(\pi^2/6 - \mathrm{Li}_2(e^{-2\pi\Delta})\big)/(2\pi^2\Delta^2)$ quoted in [8].

3. The optimal window kernel

The map $f \mapsto W_E(f)$ is a quadratic form, $W_E(f) = \mathcal{B}_E(f,f)$, where $\mathcal{B}_E$ is obtained from (1) by replacing $g = f * f$ with the symmetrised correlation of two functions. Fix the basis

$$b_k(x) = P_{2k}(x/L)\,\mathbf{1}_{[-L,L]}(x), \qquad 0 \le k < K,$$

of even polynomials on the window, where P_m is the Legendre polynomial of degree m. Let $G_{ij} = \mathcal{B}_E(b_i,b_j)$ be the Gram matrix and $v_i = \hat b_i(0) = \int b_i$, so that $v = (2L, 0, \dots, 0)$. For $f = \sum c_k b_k$ we have $W_E(f) = c^{\mathsf T} G c$ and $\hat f(0) = v^{\mathsf T} c$.

Proposition 3.1. *If G is positive definite, the minimum of $W_E(f)/\hat f(0)^2$ over nonzero f in* $\mathrm{span}(b_0,\dots,b_{K-1})$ *equals*

$$B_{\mathrm{rad}}(L) = \frac{1}{v^{\mathsf T} G^{-1} v},$$

and is attained at $c^* = G^{-1}v/(v^{\mathsf T}G^{-1}v)$.

Proof. Minimise $c^{\mathsf T}Gc$ subject to $v^{\mathsf T}c = 1$ (Lagrange multipliers, or Cauchy–Schwarz in the inner product $\langle x,y\rangle = x^{\mathsf T}Gy$). □

Since $v = \Phi(0)$, the quantity $v^{\mathsf T}G^{-1}v$ is the value at the central point of the following kernel.

Definition 3.2 (Radix Riemanniana)**.** Let $\Phi(z) = (\hat b_0(z), \dots, \hat b_{K-1}(z))^{\mathsf T}$. For an elliptic curve $E/\mathbb{Q}$ and a window $[-L,L]$, the *radix Riemanniana* of E (of order K) is the kernel

$$\mathcal{R}_{E,L}(w,z) \;=\; \Phi(w)^*\, G^{-1}\, \Phi(z),$$

that is, the reproducing kernel of the space $\{\hat f : f \in \mathrm{span}(b_k)\}$ equipped with the Weil inner product $\mathcal{B}_E$. It is built from the Gamma factor, the conductor and the primes $p < e^{2L}$ alone, without reference to the zeros of $L(E,s)$. The optimal bound of Proposition 3.1 is

$$B_{\mathrm{rad}}(L) = \frac{1}{\mathcal{R}_{E,L}(0,0)}.$$

Remark 3.3. As $K \to \infty$ the radix Riemanniana is the window analogue, for $L(E,s)$, of the reproducing kernels of the Hilbert spaces of entire functions attached to the Weil form of ζ in [14]. The name and the construction from prime data go back to the author's work on Weil positivity for ζ.

Proposition 3.4. *For every L with G positive definite, $B_{\mathrm{rad}}(L) \le B_{\mathrm{std}}(L)$.*

Proof. $\mathbf{1}_{[-L,L]} = b_0$ lies in the span, so $B_{\mathrm{std}}(L)$ is one of the values minimised over in Proposition 3.1. □

Positive definiteness of G holds under GRH except in degenerate cases, since $W_E(f) = \sum_\gamma \hat{f}(\gamma)^2$. In practice we simply attempt a Cholesky factorisation and discard the window if it fails. Note that *the certificate in Section 4 does not depend on optimality or on positive definiteness*: it evaluates $W_E(f)$ for the specific f that was found.

4. Certification

The optimisation is carried out in double precision. The resulting coefficients are converted to exact rationals $c_k \in \mathbb{Q}$, and L is taken to be an exact decimal. For this fixed $f = \sum c_k b_k$ we evaluate $W_E(f)/\hat{f}(0)^2$ rigorously in ball arithmetic (Arb [1] via `python-flint`, 256-bit precision):

(1) On $[0, 2L]$, $g = f * f$ is a polynomial with rational coefficients. We compute it exactly in $\mathbb{Q}[u]$ and obtain $\hat{f}(0)$ exactly.
(2) The archimedean integral is split at $u = 1/4$. On $[0, 1/4]$ we integrate $(g(0) - g(u))/u$ against the Bernoulli expansion $u/(e^u - 1) = \sum_n B_n u^n/n!$, truncated after 80 terms. The tail is bounded with $|B_n|/n! \le 4(2\pi)^{-n}$. On $[1/4, 2L]$ the integrand is analytic and is integrated rigorously by Arb's certified Gauss–Legendre integration. The constants γ_{E}, π and the logarithms are balls.
(3) The prime sum runs over all prime powers $n < e^{2L}$. The quantities $\log n$, $\sqrt{n}$ and c_n are balls computed from the exact integers a_p. If the ball for $\log n$ contains $2L$, the window is rejected.

The output is a ball containing $W_E(f)/\hat{f}(0)^2$. A statement $r_{\mathrm{an}}(E) < r + 1$ is recorded only if the upper end of the ball is below $r + 1$. The a_p are computed with PARI/GP [11].

5. Validation against the zeros

Identity (1) lets us check the whole pipeline (Rust optimiser → rational coefficients → certified prime side) against an independent computation. We compute the zeros γ of $L(E,s)$ up to height T with PARI's `lfunzeros` and evaluate the zero side $r_{\mathrm{an}}\hat{f}(0)^2 + \sum_{0<|\gamma|\le T} \hat{f}(\gamma)^2$ for the same certified f. Since every omitted term is nonnegative, the truncated zero side must lie below the prime side, and the difference must tend to 0 as $T \to \infty$.

For 37a1 the difference is $8.1 \cdot 10^{-3}$, $4.4 \cdot 10^{-3}$ and $2.4 \cdot 10^{-3}$ at $T = 300$, 600 and 1200 (446, 1024 and 2313 zeros). It roughly halves with each doubling of T, consistent with a truncation tail of order $(\log T)/T$.

6. Benchmarks

For each curve we scan L on a grid of step 0.05 with $K = 10$ basis functions. For both B_{std} and B_{rad} we record the first L at which the certified bound drops below $r_{\mathrm{an}} + 1$ (criterion A), and the first L at which it drops below $r_{\mathrm{an}} + 2$ (criterion B; together with the root number, this already determines the parity of r_{an}). Cost is measured as the number of primes $p < e^{2L}$ used.

curve	r_{an}	L	prime side (certified)	zero side, $\lvert\gamma\rvert \le T$	T	difference
37a1	1	0.40	1.9967148280	1.9923507969	600	$4.4 \cdot 10^{-3}$
389a1	2	0.65	2.9926821162	2.9903944027	600	$2.3 \cdot 10^{-3}$
5077a1	3	0.90	3.9514652620	3.9497072474	600	$1.8 \cdot 10^{-3}$
234446a1	4	1.15	4.9323755046	4.9292024883	300	$3.2 \cdot 10^{-3}$

TABLE 1. Certified prime side versus zero side, both normalised by $\hat{f}(0)^2$, for the optimised test function at the cutoff where the bound first drops below $r_{\mathrm{an}} + 1$.

Because of the grid, prime counts at the threshold are quantised. Ties are therefore frequent and slightly understate the gain.

6.1. **Cremona's tables.** We take one curve per isogeny class, sampled up to 150 per rank from three blocks of Cremona's tables [5] ($61 \le N \le 409989$, 1642 curves, ranks 0–3), with the cutoff $L \le 4.5$ (1028 primes). Table 2 reports the results. In total 6449 bounds were certified, and none lies below the known rank.

	curves decided (std / rad)	rad fewer / equal / more	primes saved
criterion A ($< r_{\mathrm{an}} + 1$)	1573 / 1592	578 / 1064 / 0	15.2%
criterion B ($< r_{\mathrm{an}} + 2$)	1642 / 1642	263 / 1379 / 0	8.3%

criterion A, by rank	0	1	2	3
curves (both decided)	381	450	450	292
primes saved	18.8%	4.3%	3.9%	4.5%

TABLE 2. Cremona sample. "Primes saved" is $1 - \sum P_{\mathrm{rad}} / \sum P_{\mathrm{std}}$ over curves decided by both methods.

6.2. **The Stein–Watkins database.** From the Stein–Watkins database [13] (files of the 2011 distribution) we sample 250 curves per (analytic) rank $0, 1, 2, 3, \ge 4$ from two groups: general conductors $4.4 \cdot 10^6 \le N \le 9.7 \cdot 10^7$ and prime conductors $3.6 \cdot 10^9 \le N \le 7.5 \cdot 10^9$, giving 2500 curves in total. We check every conductor and root number with PARI. The ranks in the database are Watkins' numerically determined analytic ranks. The cutoff is $L \le 5.75$ (9592 primes). In total 9105 bounds were certified, and none contradicts the database rank.

	decided (std / rad)	rad fewer / equal / more	primes saved
criterion A	2059 / 2096	1306 / 1194 / 0	17.0%
criterion B	2466 / 2484	889 / 1611 / 0	22.7%

criterion B, by rank	0	1	2	3	≥ 4
curves (both decided)	466	500	500	500	500
primes saved	24.7%	15.1%	5.7%	3.1%	2.7%

TABLE 3. Stein–Watkins sample.

6.3. **Large conductors and high rank.** Table 4 lists every curve of an additional test set, including the cases without gain. Prime counts up to e^{16} were handled by streaming the prime sum.

curve	N	rank	criterion	primes std / rad	saved
twist of 11a1, $d = 10009$	1101980891	1	B	329 / 302	8%
twist of 389a1, $d = 10009$	38970051509	1	B	3523 / 3523	0%
twist of 11a1, $d = 100049$	110107826411	0	B	2466 / 2065	16%
twist of 5077a1, $d = 10009$	508614271237	1	B	1220 / 1019	16%
twist of 389a1, $d = 100049$	3893813133989	0*	B	17885 / 13613	24%
twist of 5077a1, $d = 100049$	50819766789877	1	B	1220 / 1220	0%
twist of 389a1, $d = 1000033$	389025674423621	0*	B	102084 / 77395	24%
twist of 5077a1, $d = 1000033$	5077335087528853	1	B	340436 / 195040	43%
twist of 11a1, $d = 1000033$	11000726011979	0	B	undecided by both for $L \le 7.5$	
5187563742a1	5187563742	6	A	11 / 10	9%
$[1, -1, 0, -106384, 13075804]$	249649566346838	8	A	21 / 18	14%
$[1, -1, 0, -135004, 97151644]$	32107342006814614	9	A	32 / 29	9%

TABLE 4. Additional curves. Ranks are lower bounds from PARI's `ellrank` (points). *Here the 2-descent leaves rank 0 or 2 open. The root number $+1$ together with the certified bound $r_{\mathrm{an}} < 2$ gives $r_{\mathrm{an}} = 0$ under GRH, and hence rank 0 by Kolyvagin [9].

7. Validation against published fixed-kernel computations

To make sure that B_{std} is *exactly* the method of [2, 8], and not a variant of it, we re-implemented Bober's formula [2, eq. (3)] directly from the paper, with the archimedean term in Spicer's closed form and without any code shared with our optimiser. Conductors were taken from the ICARM leaderboard [7]; $\log N$ agrees with [2, Table 1] to all printed digits. Table 5 shows that the independent implementation reproduces every published value we could find. The floating-point $B_{\mathrm{std}}(\pi\Delta)$ of our optimiser agrees with it to $2 \cdot 10^{-7}$, and the Arb certificate to better than 10^{-12}.

curve	$\log N$	Δ	published	independent	certified $B_{\mathrm{std}}(\pi\Delta)$
E_{20}	170.09	2.0	21.70 [2]	21.690724	$[21.6907238431 \pm 5 \cdot 10^{-13}]$
E_{21}	196.68	2.5	22.68 [2]	22.672713	
E_{22}	182.72	2.0	23.71 [2]	23.704751	$[23.7047512825 \pm 3 \cdot 10^{-13}]$
E_{23}	205.06	2.5	24.49 [2]	24.483411	
E_{24}	219.93	2.5	25.57 [2]	25.568173	$[25.5681730272 \pm 5 \cdot 10^{-13}]$
E_{27}	287.01	3.1	< 29 [8]	28.988058	
E_{28}	325.90	3.2	31.30 [2]	31.298359	

TABLE 5. Reproduction of published values of $\sum_\gamma f(\gamma;\Delta)$; [2] rounds up. E_{20}–E_{24}, E_{28} are the curves of [2], E_{27} that of [8].

8. Curves of record rank

For curves of large rank the cutoff, and with it the number of primes, grows quickly: Klagsbrun, Sherman and Weigandt used $\Delta = 4$ (primes up to $e^{8\pi} \approx 8 \cdot 10^{10}$, about 40 hours) for the Elkies curve [8], and the leaderboard record of rank 30 was certified with $\Delta = 4.25$ (primes up to $\approx 4 \cdot 10^{11}$) [7]. This is where a better test function matters most. The following comparison was suggested by Z. Klagsbrun.

Setup. All curves are taken from the ICARM Elliptic Curve Rank Leaderboard [7] (snapshot of 12 October 2026, 4201 curves) in their global minimal models; the rank r is the leaderboard's certified lower bound. We recomputed every conductor from local data with PARI and checked that the root number equals $(-1)^r$ in every case, so that a certified bound $B < r + 2$ gives $r_{\mathrm{an}} \le r$ under GRH (and rank exactly r under GRH and BSD). We computed a_p for all $p < 10^9$ (50 847 534 primes), scanned L with step 0.01 and $K = 10$ (for E_{28}, $K = 14$ and $K = 20$ give the same values to 10^{-4}), and certified every reported value with Arb as in Section 4.

Ranks 28–31. For each rank 28–31 we take the curve of smallest conductor. Table 6 lists the number of primes needed. In all four cases the optimised kernel reaches each target with a window about 1 shorter, which reduces the prime bound by roughly e^2. Near $r + 1$ the fixed-kernel bound flattens, and the gap becomes much larger (Figure 1).

			bound $< r+2$		bound $< r+1$	
ICARM	r	$\log N$	fixed	optimal	fixed	optimal
#3762	28	242.9	122 776	**19 235**	21 600 954	**234 855**
#3835	29	265.4	316 068	**48 890**	$> 5.08 \cdot 10^7$ [a]	**717 477**
#3897	30	279.9	387 811	**59 820**	$> 5.08 \cdot 10^7$ [a]	**691 123**
#3900	31	325.3	4 340 969	**665 760**	$> 5.08 \cdot 10^7$ [a]	**13 209 786**

Table 6. Number of primes $p < e^{2L}$ at the first certified L (grid step 0.01). [a]Not reached with all primes $p < 10^9$; the certified fixed-kernel bounds at $L = 10.36$ are $[30.00290409 \pm 3.2 \cdot 10^{-9}]$, $[31.08984065 \pm 3.1 \cdot 10^{-9}]$ and $[32.43064260 \pm 2.5 \cdot 10^{-10}]$.

The historic record curves. On the record curves of [2, 8] the gain is smaller but consistent (Table 7). For E_{27} the minimal fixed-kernel window $L = 9.72$ ($\Delta = 3.09$) matches the choice $\Delta = 3.1$ of [8]. For E_{28} neither kernel reaches a bound below 30 with $p < 10^9$ (at $L = 10.36$: 31.092 fixed, 30.820 optimal).

	E_{20}	E_{21}	E_{22}	E_{23}	E_{24}	E_{27}
fixed, $< r+2$	11 988	181 084	11 988	77 395	103 973	15 078 540
optimal, $< r+2$	**5 515**	**83 298**	**5 133**	**23 068**	**32 069**	**4 340 969**
optimal, $< r+1$	47 114	—	37 128	353 398	433 671	—

Table 7. Historic record curves, primes needed (grid step 0.01, all values certified). The fixed kernel reaches no bound below $r + 1$ for $p < 6.7 \cdot 10^6$.

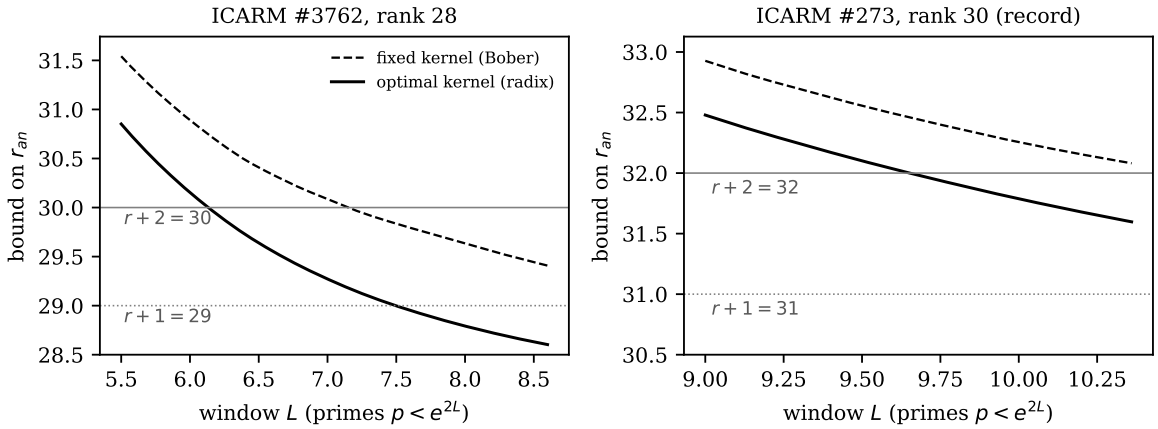


FIGURE 1. Fixed-kernel and optimal-kernel bounds as functions of the window L (floating point, grid step 0.01).

The 2026 records. In August 2026 curves of rank at least 30 (#273) and 31 (#302) were found by Claude with L. Alpöge and A. Howell [7]. For #273 ($\log N = 339.35$, root number $+1$) the leaderboard reports a fixed-kernel bound below 32 with $\Delta = 4.25$. The optimised kernel certifies

$$r_{\mathrm{an}}(\#273) \;\le\; [31.99478129 \pm 4.5 \cdot 10^{-9}] \;<\; 32$$

at $L = 9.66$, with 13 461 681 primes $p < 2.4 \cdot 10^8$. The fixed kernel does not reach 32 with all primes below 10^9: its certified value at $L = 10.36$ is $[32.08095018 \pm 1.5 \cdot 10^{-9}]$. Hence the optimised kernel needs fewer than a quarter of the primes the fixed kernel needs at the very least, and about 1/1100 of the roughly $1.5 \cdot 10^{10}$ primes below $e^{8.5\pi}$. For #302 ($\log N = 375.22$, root number -1) neither kernel reaches 33 with $p < 10^9$ (at $L = 10.36$: fixed $[34.02864458 \pm 1.3 \cdot 10^{-9}]$, optimal 33.566). A fit of the form $A + C/L$ suggests $p \lesssim 6 \cdot 10^9$ for the optimal and $p \lesssim 3 \cdot 10^{10}$ for the fixed kernel.

9. LIMITATIONS

- All bounds assume GRH for $L(E, s)$ and concern the analytic rank. Statements about the algebraic rank additionally use points, the root number and, for $r_{\mathrm{an}} \le 1$, the theorems of Gross–Zagier [6] and Kolyvagin [9].
- The basis size is fixed at $K = 10$ and the cutoff grid at step 0.05 (0.01 in Sections 7 and 8). Larger K can only improve B_{rad}, and a finer grid would turn some ties into gains. We have not optimised either choice.
- The certification is currently performed in Python on top of Arb. The optimiser is written in Rust.
- The gain depends on the curve: on record-rank curves it ranges from a factor of about 2 (E_{20}, E_{21}) to 6.5 (ICARM, ranks 28–31) for a bound below $r + 2$. For E_{28} and #302 we could not reach the target with $p < 10^9$ and report only extrapolations.
- Ranks in Section 8 are certified lower bounds from the leaderboard. Our bounds concern the analytic rank; equality with the algebraic rank requires BSD.

Data and code availability

Code (Rust optimiser, certification scripts), the sampled curve lists and all certificates are available at `https://github.com/sectio-aurea-q/radix-rank` (archived on Zenodo, DOI: 10.5281/zenodo.23299461). The folders `validation/` and `highrank/` contain the independent implementation of Bober's formula and the scripts and certified values for Sections 7 and 8. Running `./reproduce.sh quick` re-derives and re-certifies the bounds for seven reference curves in about one minute.

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